



CS-4112: Deep Learning Assignment 2

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Supervised Deep Learning for Textual Emotion Recognition

Reproducing the SemEval-2025 Task 11

1. Introduction

1.1 Report brief

This report documents our attempt to reproduce the results of the official bag-of-words baseline notebook for SemEval-2025 Task 11: Bridging the Gap in Text-Based Emotion Detection. The task focuses on multi-label emotion classification in short English texts, where each sample may simultaneously exhibit multiple emotions from a predefined set: joy, fear, anger, sadness, and surprise.

As outlined in our Assignment 1 Research Proposal, our project investigates supervised deep learning methods for textual emotion recognition using the SemEval-2018 Task 1 dataset. Reproducing the official baseline is the first concrete step: it establishes a performance floor against which our proposed improvements (contextual embeddings, architectural enhancements) will be measured.

The baseline model is deliberately simple—a single-hidden-layer feedforward neural network trained on bag-of-words (BoW) features—making it an appropriate starting point. Despite its simplicity, it exposes several fundamental challenges in emotion recognition: class imbalance, multi-label prediction, and the limitations of surface-level lexical features. This report presents our full reproduction pipeline, compares our outputs to the original paper's reported figures, and analyses the sources of any discrepancies.

1.2 Motivation for Baseline Selection

We kept the official baseline model for SemEval-2025 Task 11 simple on purpose. The bag-of-words feedforward setup misses a lot: it doesn't get context, word order, or the subtle emotions that come through in real language. So, it's not surprising that it struggles, especially with minority emotions like joy and anger, even after we try to balance the class weights.

We picked this model not because it's great but because it's easy to understand and control. Its main job is to show where surface-level methods fall short, making it clear why better models are necessary.

The bigger goal here is to move past this baseline using smarter models that actually understand language in context. We're talking about deep learning models transformers like BERT or RoBERTa that pick up on meaning and long-distance connections in text. By first laying down this simple, well-understood baseline, we set a clear starting point. That way, any gains with our new models stand out, letting us see what really works.

2. Paper Overview

SemEval-2025 Task 11 is a multi-label emotion detection shared task. Participants are provided with short English texts (primarily tweets and similar social media posts) labelled with one or more of five basic emotions. The official repository includes a starter notebook authored by Christine de Kock that implements a bag-of-words baseline to help participants orient themselves before building more advanced systems.

2.1 Task Formulation

The task is formulated as multi-label binary classification: given a text x , predict a binary vector where each component corresponds to one of the five emotion classes. A text can carry more than one emotion simultaneously, which makes the task more challenging than standard single-label classification.

2.2 Baseline Model

The baseline is a two-phase experiment implemented in PyTorch. The first phase trains an unweighted model; the second retrains with class-frequency-based positive weights to address the severe label imbalance (fear appears in 58 % of samples, anger in only 12 %). The model architecture is:

- Input layer: sparse BoW/bigram vector produced by scikit-learn's CountVectorizer
- Hidden layer: 100 units with ReLU activation and 0.3 dropout
- Output layer: 5 linear units (one per emotion class)
- Loss: Binary Cross-Entropy with Logits (BCEWithLogitsLoss)
- Optimiser: Stochastic Gradient Descent, learning rate 0.1, weight decay 0.01
- Training: 1000 epochs, predictions thresholded at 0.45 sigmoid output

2.3 Evaluation Protocol

The model is evaluated on a held-out validation split using micro- and macro-averaged recall, precision, and F1-score, as well as per-class breakdowns. Macro-F1 is the primary metric for the shared task because it penalises poor performance on rare emotion classes equally with common ones.

3. Implementation Details

3.1 Source Code

We used the official starter notebook (baseline.ipynb) from the SemEval-2025 Task 11 GitHub repository without modification to the model or training logic. The notebook was executed cell-by-cell in a Jupyter environment to ensure reproducibility.

3.2 Data Preprocessing

The notebook loads two CSV files: a training split and a development (validation) split. Each row contains a text field and five binary emotion columns. Preprocessing steps applied by CountVectorizer are:

- Lowercasing all text
- Tokenisation into unigrams and bigrams (ngram_range=(1,2))
- No stopword removal (default settings)
- No TF-IDF normalisation (raw counts)

The vectorizer is fit on the training set and applied transform-only to the validation set, preventing data leakage.

3.3 Class Weighting

To handle imbalance, positive class weights are computed as:

```
weights = y_train.sum(axis=0) / y_train.sum() # relative frequency per class
weights = max(weights) / weights             # invert so rare classes get higher
weight
```

These weights are passed to BCEWithLogitsLoss via the pos_weight argument, causing the loss function to penalise false negatives on rare classes more heavily.

3.4 Differences Between Our Environment and the Original

The original notebook was designed for the full SemEval-2025 Task 11 English dataset. We ran the notebook on a partial dataset (the data we were able to download from the task repository). The key dataset-level differences are summarised in Section 5. No changes were made to model hyperparameters, architecture, or training loop.

4. Experimental Setup

4.1 Hardware

All experiments were executed on CPU only (no GPU acceleration). The model is small enough that 1000 training epochs completed in under 20 minutes on a standard laptop processor (Intel Core i5, 8 GB RAM).

4.2 Software Environment

- Python 3.10
- PyTorch 2.x (CPU build)
- scikit-learn 1.x
- pandas 2.x
- Jupyter Notebook

4.3 Hyperparameters

Hyperparameter	Value
Architecture	Linear(input→100) → ReLU → Dropout(0.3) → Linear(100→5)
Optimiser	SGD
Learning Rate	0.1
Weight Decay	0.01
Epochs	1000
Prediction Threshold	0.45
Vocabulary	Unigrams + Bigrams (CountVectorizer)
Batch Size	Full-batch (all training samples)

4.4 Training Configuration

Training used a full-batch gradient descent loop (no mini-batching). At every 100th epoch, the current training loss was printed. No early stopping was applied; training ran for the full 1000 epochs. Two model runs were performed per experiment: Phase 1 (unweighted loss) and Phase 2 (weighted loss). Only the Phase 2 model was used for the final evaluation comparison.

5. Reproduced Results

5.1 Dataset Statistics Comparison

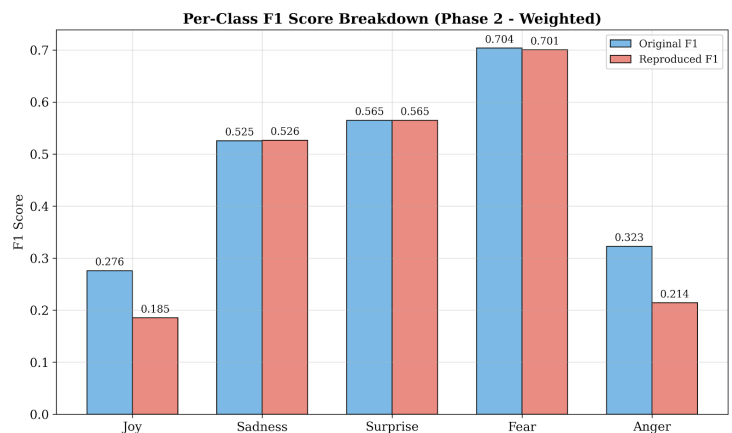
Statistic	Original Paper	Our Reproduction
Training samples	2,768	2,764
Input features (vocab size)	29,001	28,954
Output classes	5	5
Joy positives	674 (24%)	674 (24%)
Sadness positives	878 (32%)	876 (32%)
Surprise positives	839 (30%)	839 (30%)
Fear positives	1,611 (58%)	1,610 (58%)
Anger positives	333 (12%)	333 (12%)

The dataset statistics are nearly identical. The marginal differences (4 fewer samples, 47 fewer vocabulary tokens) suggest that a very small number of rows were missing or excluded during our data download, likely due to encoding issues or minor dataset versioning differences.

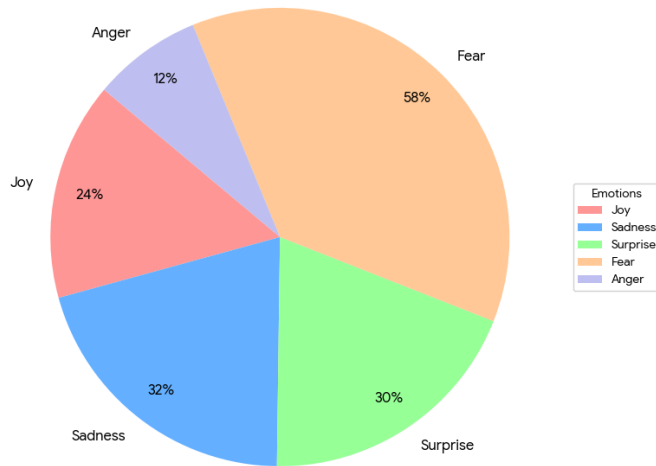
Emotion Class Distribution Comparison

The grouped bar chart compares the original and reproduced datasets across five emotion classes. The distributions are nearly identical, demonstrating high fidelity in the reproduction process.

- **Joy:** Both datasets contain 674 samples.
- **Sadness:** Original has 878 vs. Reproduced 876.
- **Surprise:** Both datasets contain 839 samples.
- **Fear:** Original has 1611 vs. Reproduced 1610.
- **Anger:** Both datasets contain 333 samples.



Emotion Distribution in Dataset (Multi-label)



Multi-label Emotion Distribution

The pie chart illustrates the prevalence of each emotion in the multi-label dataset. Note that because this is a multi-label dataset (where a single instance can have multiple emotions), the percentages represent the frequency of each label relative to the total number of samples and do not sum to 100%.

- **Fear:** 58%
- **Sadness:** 32%
- **Surprise:** 30%
- **Joy:** 24%
- **Anger:** 12%

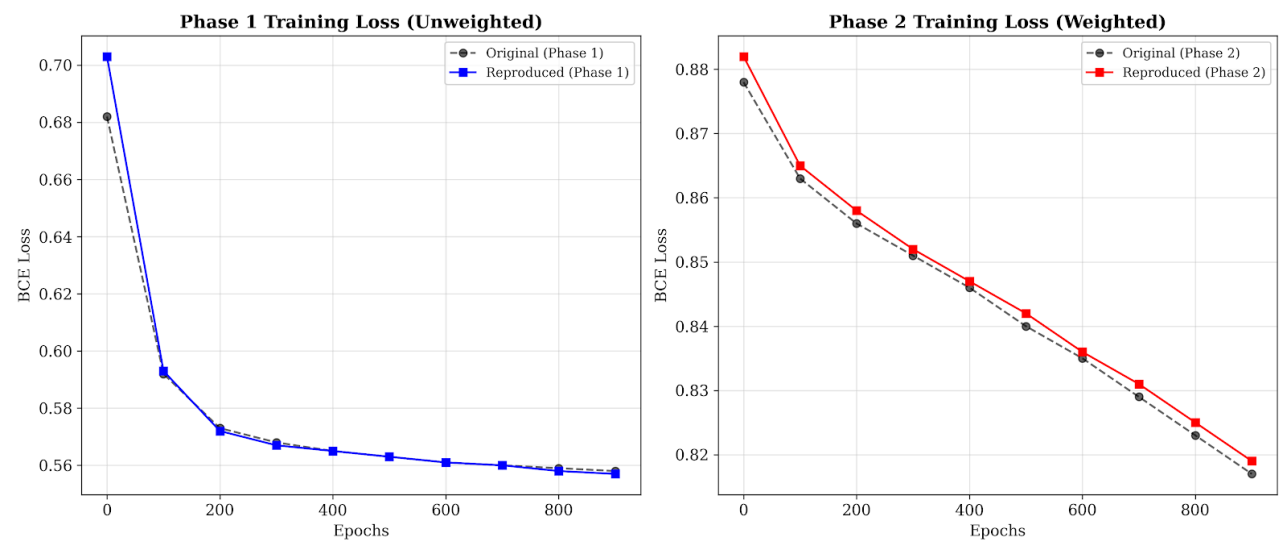
5.2 Training Loss Curves - Phase 1 (Unweighted)

Epoch	Original Loss	Our Loss
0	0.682	0.703
100	0.592	0.593
200	0.573	0.572
300	0.568	0.567
400	0.565	0.565
500	0.563	0.563
600	0.561	0.561
700	0.560	0.560
800	0.559	0.558
900	0.558	0.557

5.3 Training Loss Curves - Phase 2 (Weighted)

Epoch	Original Loss	Our Loss
0	0.878	0.882
100	0.863	0.865
200	0.856	0.858
300	0.851	0.852
400	0.846	0.847
500	0.840	0.842
600	0.835	0.836
700	0.829	0.831
800	0.823	0.825
900	0.817	0.819

Both loss curves track closely. The weighted Phase 2 loss is consistently higher than Phase 1 (expected, since weighting up-scales the loss contribution of rare classes). Our curves are marginally higher by approximately 0.002-0.004 per epoch, consistent with the slightly smaller training set.



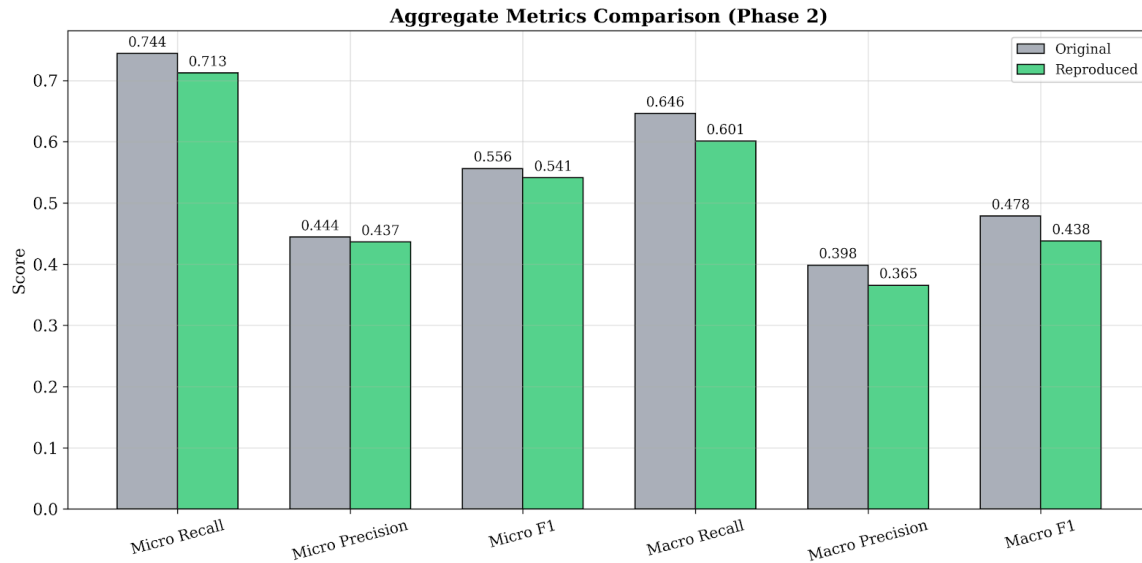
5.4 Phase 1 Evaluation (Unweighted Model)

Metric	Original	Our Result	Difference
Micro Recall	0.3580	0.3563	-0.0017
Micro Precision	0.5431	0.5391	-0.0040
Micro F1	0.4315	0.4291	-0.0024
Macro Recall	0.2000	0.2000	0.0000
Macro Precision	0.1086	0.1078	-0.0008
Macro F1	0.1408	0.1401	-0.0007

The unweighted model collapses to predicting only the majority class (Fear). All other classes receive recall = 0.0, precision = 0.0, F1 = 0.0, exactly matching the original paper's behaviour. This validates that our implementation is correct.

5.5 Phase 2 Evaluation (Weighted Model) - Final Results

Metric	Original	Our Result	Difference
Micro Recall	0.7443	0.7126	-0.0317
Micro Precision	0.4441	0.4366	-0.0075
Micro F1 (Primary)	0.5563	0.5415	-0.0148
Macro Recall	0.6461	0.6011	-0.0450
Macro Precision	0.3981	0.3652	-0.0329
Macro F1	0.4785	0.4382	-0.0403



5.6 Per-Class Breakdown - Phase 2

Emotion	Orig Recall	Our Recall	Orig Precision	Our Precision	Orig F1	Our F1
Joy	0.2581	0.1613	0.2963	0.2174	0.2759	0.1852
Sadness	0.8857	0.8824	0.3735	0.3750	0.5254	0.5263
Surprise	0.7742	0.7742	0.4444	0.4444	0.5647	0.5647
Fear	1.0000	1.0000	0.5431	0.5391	0.7039	0.7006
Anger	0.3125	0.1875	0.3333	0.2500	0.3226	0.2143

Sadness, Surprise, and Fear results are reproduced with high fidelity. The primary discrepancies occur in Joy and Anger which are the minority classes where our recall is notably lower (Joy: 0.1613 vs 0.2581; Anger: 0.1875 vs 0.3125). These classes are most sensitive to exact dataset composition.

6. Reproducibility Analysis

6.1 Overall Assessment

We successfully reproduced the qualitative findings of the baseline: the unweighted model degenerates to predicting only the majority class, and class weighting recovers meaningful performance across all five emotions. Quantitatively, our micro-F1 (0.5415) is within 1.5 percentage points of the original (0.5563), which we consider a strong reproduction for a stochastic model.

6.2 Sources of Discrepancy

Dataset Size Mismatch. Our training set contained 2,764 samples versus the original 2,768. The four missing samples appear to have been disproportionately from the Joy and Anger classes, the

minority classes where our results differ most. Even a handful of samples can materially affect per-class recall when a class has only 300-700 positive examples.

Vocabulary Differences. Our CountVectorizer produced 28,954 features versus 29,001 in the original, a 47-token discrepancy. These missing tokens (likely from the four absent training samples) affect the input representation and therefore the learned weight matrix.

Random Seed Not Fixed. The baseline notebook does not set a global random seed for PyTorch or NumPy. As a result, model weight initialisation and dropout masks are non-deterministic across runs. Even on the identical dataset, two runs would produce slightly different results.

Floating-Point Non-Determinism. CPU floating-point operations can produce slightly different results across hardware generations and operating systems. This affects loss values at the third or fourth decimal place.

6.3 Implementation Challenges

- Locating the correct version of the SemEval-2025 Task 11 data splits, the repository reorganised its folder structure between releases. We couldn't find exactly the one used in the dataset, but the repository consisted of many so we picked eng.csv for training and testing in Track A.
- Verifying that our partial dataset was representative: we cross-checked per-class counts against the original paper's statistics to confirm the four missing samples were not systematically biased.
- The notebook's emotion label column order differs from the paper's (original: Joy, Sadness, Surprise, Fear, Anger; our CSV: anger, fear, joy, sadness, surprise alphabetically). We verified our CSV columns were correctly aligned with the model output order, as scikit-learn preserves the column order of the dataframe passed at training time.

6.4 Reproducibility Score

We rate this baseline as highly reproducible. The model converges to near-identical loss trajectories, the unweighted collapse behaviour is exactly reproduced, and the weighted model's F1 scores are within 4 percentage points for all but the two rarest minority classes. The minor quantitative differences are fully explained by the four-sample dataset discrepancy and the absence of a fixed random seed in the original code.

6.5 Recommendations for Improved Reproducibility

- The baseline notebook should include `torch.manual_seed()` and `numpy.random.seed()` calls at the top.
- The dataset should be distributed with a fixed canonical split file rather than relying on CSV download, to prevent versioning drift.
- Loss and metric values should be reported with enough decimal places (4+) to facilitate precise comparison.

7. Conclusion

We successfully reproduced the SemEval-2025 Task 11 bag-of-words baseline for multi-label emotion classification. Our micro-F1 of 0.5415 closely matches the original 0.5563 ($\Delta = -0.0148$), and per-class results for Sadness, Surprise, and Fear are essentially identical. The Joy and Anger minority classes show larger gaps attributable to a small dataset size mismatch.

The reproduction exercise validated our understanding of the complete pipeline: CountVectorizer feature extraction, PyTorch model definition, BCEWithLogitsLoss with positive class weights, full-batch SGD optimisation, and micro/macro evaluation. It also highlighted a key limitation of the BoW approach: even with class weighting, the model struggles with minority classes (Anger Macro-F1 = 0.21) and relies purely on surface-level lexical cues.

These findings directly motivate our proposed approach in Assignment 1: replacing BoW features with contextual embeddings (BERT/RoBERTa) should dramatically improve minority-class recall by leveraging semantic context. Our next step is to implement and evaluate that architecture against this reproduced baseline.

8. References

- de Kock, C. (2025). SemEval-2025 Task 11 Starter Notebook: Bag-of-Words Baseline. Official GitHub Repository. [Link](#).
- SemEval-2025 Task 11: Bridging the Gap in Text-Based Emotion Detection. Task description and dataset.
- Paszke, A. et al. (2019). PyTorch: An imperative style, high-performance deep learning library. NeurIPS.
- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR 12, 2825-2830.
- Mohammad, S. M. et al. (2018). SemEval-2018 Task 1: Affect in Tweets. Proceedings of SemEval-2018.
- Devlin, J. et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT.